

DSA 210 - Term Project Final Report

Performance Analysis of Turkish Investment Funds
vs. Risk-Free Deposit Rates (2021-2025)

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1. Motivation

In Türkiye, especially between 2021 and 2025, inflation went through the roof and the central bank kept raising interest rates. At some point deposit rates hit almost 50%. This made me wonder: if you could just put your money in a bank account and earn that much, is there even a point in investing in funds?

Everyone around me was talking about gold funds, stock funds, participation funds etc. but nobody really checked the numbers. So I wanted to answer a simple question: **which fund categories actually beat the bank deposit, and which ones are just hype?**

This project looks at all TEFAS funds over 5 years and compares each category against the CBRT deposit rate to see which ones actually delivered.

2. Data Source

TEFAS Fund Data

- Source: TEFAS (Türkiye Elektronik Fon Alım Satım Platformu)
- 1,354,763 daily observations across 2,006 funds
- Period: April 2021 - December 2025
- Fields: date, fund code, fund name, price, outstanding shares, investor count, total fund value

CBRT Deposit Rates

- Source: EVDS (Electronic Data Delivery System, Central Bank of Turkey)
- Series: TP.TRY.MT02 - weighted average TL deposit rate (commercial banks)
- 262 weekly observations, April 2021 - April 2026

The enrichment here is combining the fund performance data with the macroeconomic deposit rate data. I converted the weekly deposit rate to a daily rate ($APR / 365$) so I could calculate excess returns for each fund on each day.

3. Data Analysis

3.1 Categorisation

TEFAS does not give a clean category field, so I classified each fund by looking for keywords in the fund name. For example, if the name contains 'HISSE' it is Equity, 'ALTIN' means Gold, 'SERBEST' means Hedge, etc. This gave me 11 categories:

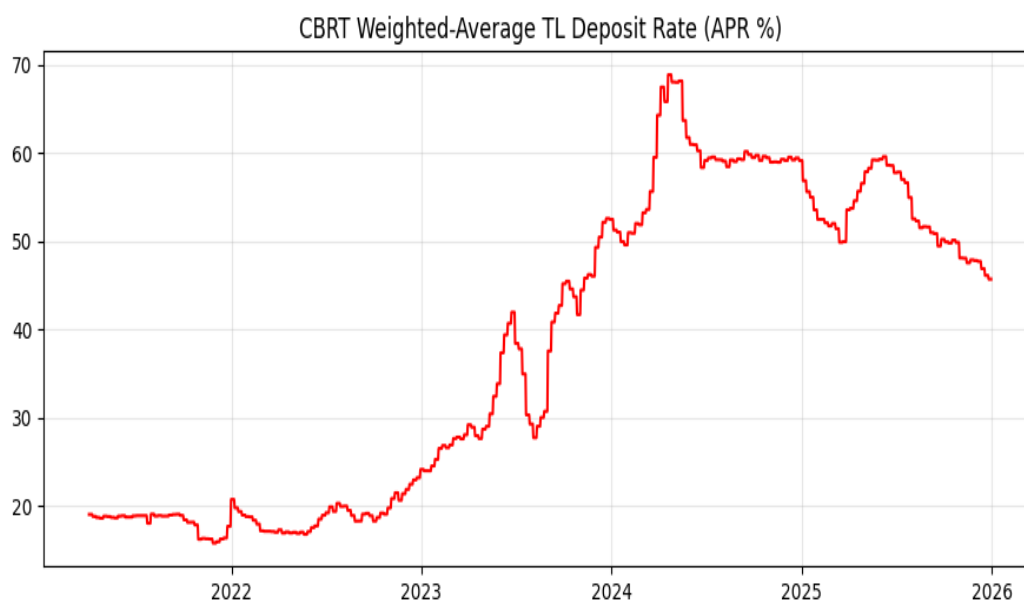
Category	# of Funds
Hedge	867
Equity	395
Variable	146
Participation	132
Money Market	120
Fund of Funds	86
Debt	74
Gold	60
Eurobond	9
Mixed	7
Other	2

3.2 Daily Returns and Excess Returns

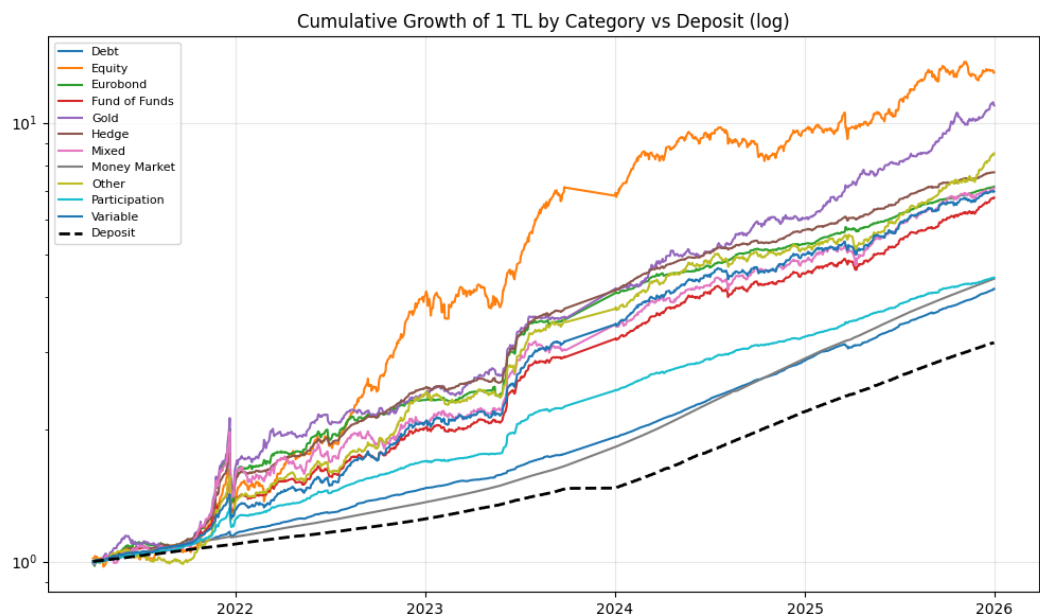
For each fund I calculated **daily return** (percentage change in price) and **excess return** (daily return minus the daily deposit rate). If excess return is positive, the fund beat the bank that day. I dropped any returns bigger than 50% since those are almost certainly data errors.

3.3 Exploratory Data Analysis

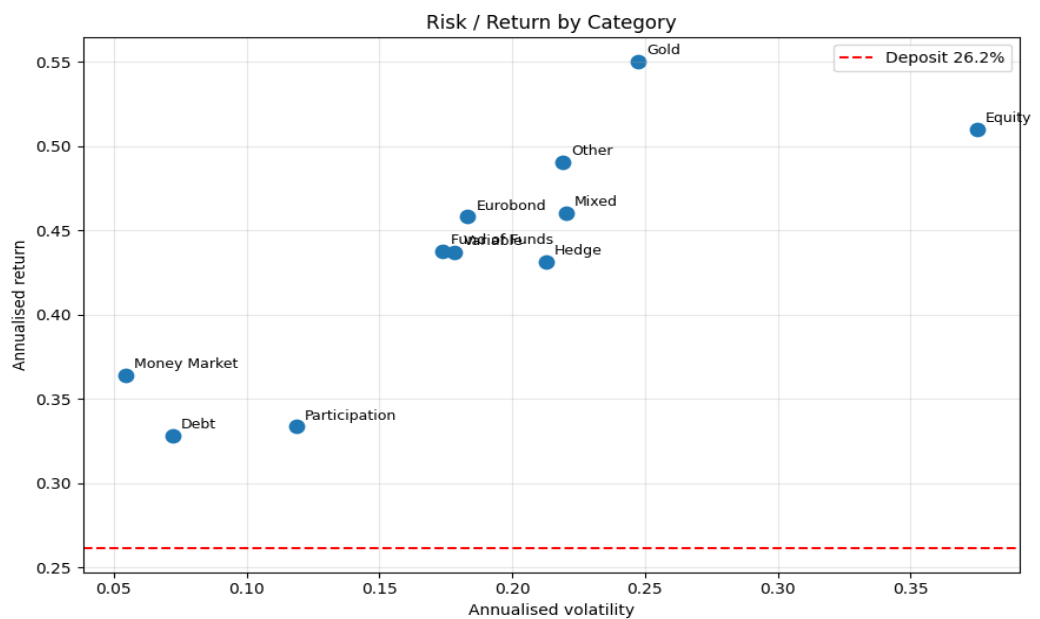
Deposit rate over time: The rate started around 19% in 2021, dropped briefly during the low-rate policy period, then shot up to almost 49% by early 2026 after the policy shift.



Cumulative growth of 1 TL: If you put 1 TL into each category in April 2021, Gold funds grew the most. The black dashed line shows the deposit benchmark.



Risk/return scatter: This was the most interesting plot for me. Money Market funds sit in the bottom-left corner (low return, very low risk). Gold and Equity are top-right (high return, high volatility). When you adjust for risk using the Sharpe ratio, Money Market is actually the best, which surprised me.



Annualised summary table:

Category	Ann. Return	Ann. Vol	Sharpe
Money Market	36.4%	5.5%	1.87
Gold	55.0%	24.7%	1.17
Eurobond	45.8%	18.3%	1.07

Category	Ann. Return	Ann. Vol	Sharpe
Fund of Funds	43.8%	17.4%	1.01
Variable	43.7%	17.8%	0.98
Debt	32.8%	7.2%	0.92
Mixed	46.0%	22.0%	0.90
Hedge	43.2%	21.3%	0.80
Equity	51.0%	37.5%	0.66
Participation	33.3%	11.9%	0.61

The average deposit rate over the period was about 26.2% per year.

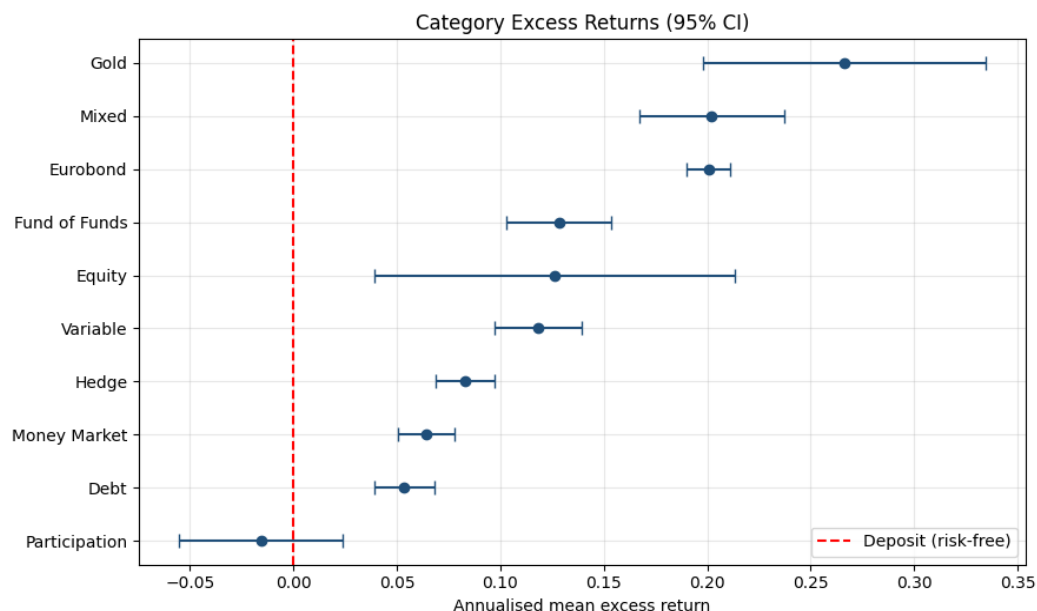
3.4 Hypothesis Testing

Test 1: One-sample t-test (per category)

- H_0 : The category's mean excess return = 0 (same as deposit)
- H_1 : The mean excess return is not 0

For each category, I took each fund's average daily excess return as one observation (to avoid autocorrelation issues with daily data). I required at least 60 days of data per fund.

Results: **9 out of 10 categories rejected H_0** - meaning they significantly beat the deposit. The only exception was **Participation (Katilim) funds**, which had a slightly negative excess return (-1.6% annualised) and failed to reject H_0 ($p = 0.437$).



Test 2: Chi-square test of independence

- H_0 : Fund category and whether a fund beats the deposit are independent
- H_1 : They are dependent

I built a contingency table where each fund is labelled 'Beat' or 'NotBeat' based on whether its average excess return was positive:

Category	Beat	NotBeat
Debt	64	10
Equity	253	142
Fund of Funds	75	11
Gold	55	5
Hedge	630	237
Money Market	119	1
Participation	82	50
Variable	118	28

Chi-square = 97.93, df = 7, $p = 3 \times 10^{-18}$. All expected counts were at least 5 (checked the assumption). **Result: REJECT H_0** - category and performance are strongly dependent.

Some observations: Money Market has 119 funds beating and only 1 not beating - almost perfect. Gold is 55 vs 5. But Participation is 82 vs 50 and Equity is 253 vs 142, much closer to a coin flip.

4. Machine Learning

I wanted to see if I could predict whether a fund will beat the deposit rate **next month** based on its current month's data.

4.1 Feature Engineering

For each fund in each month, I calculated these features:

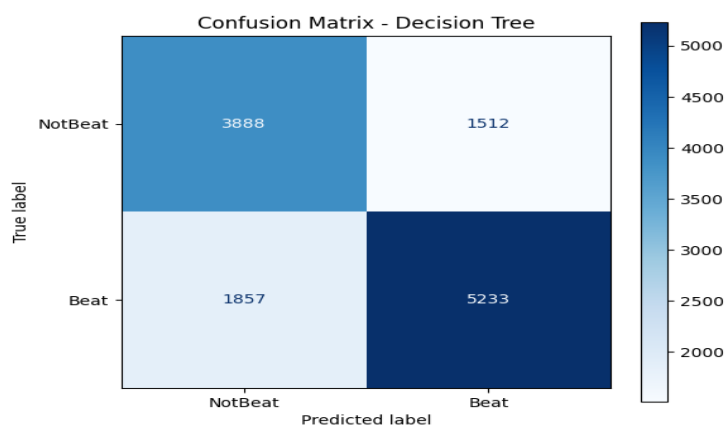
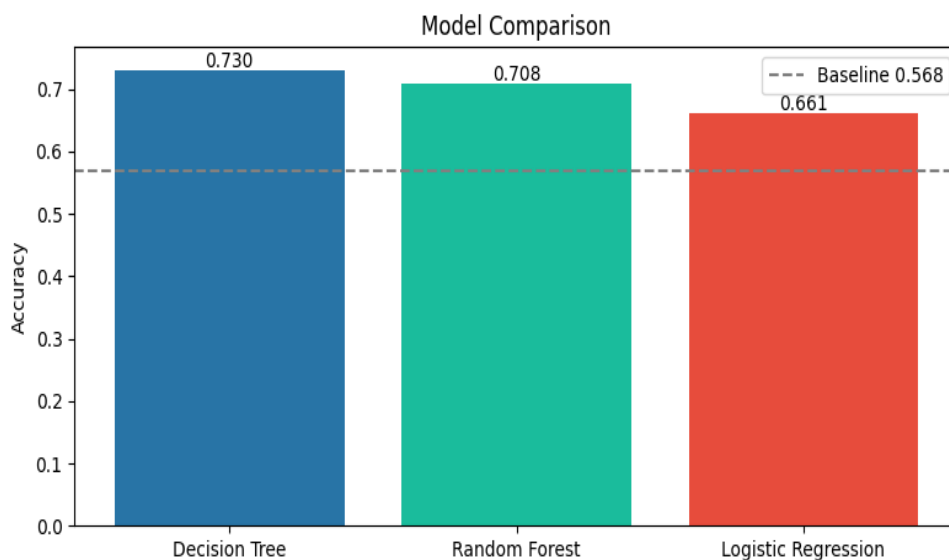
- MeanReturn - average daily return that month
- Volatility - standard deviation of daily returns (risk)
- MeanExcess - average excess return over deposit
- DepositRate - current deposit rate level
- InvestorGrowth - change in investor count
- ValueGrowth - change in total fund value
- TradingDays - number of active trading days
- Category - fund category encoded as a number

Target: did the fund beat the deposit in the *next* month? (1 = beat, 0 = not beat). This gave me 62,450 monthly observations. I split 80/20 into train and test.

4.2 Results

Model	Accuracy
Decision Tree	73.0%
Random Forest	70.8%
Logistic Regression	66.1%
Baseline (majority class)	56.8%

All three models beat the baseline, which means the features carry some predictive power. Decision Tree was the best at 73%.

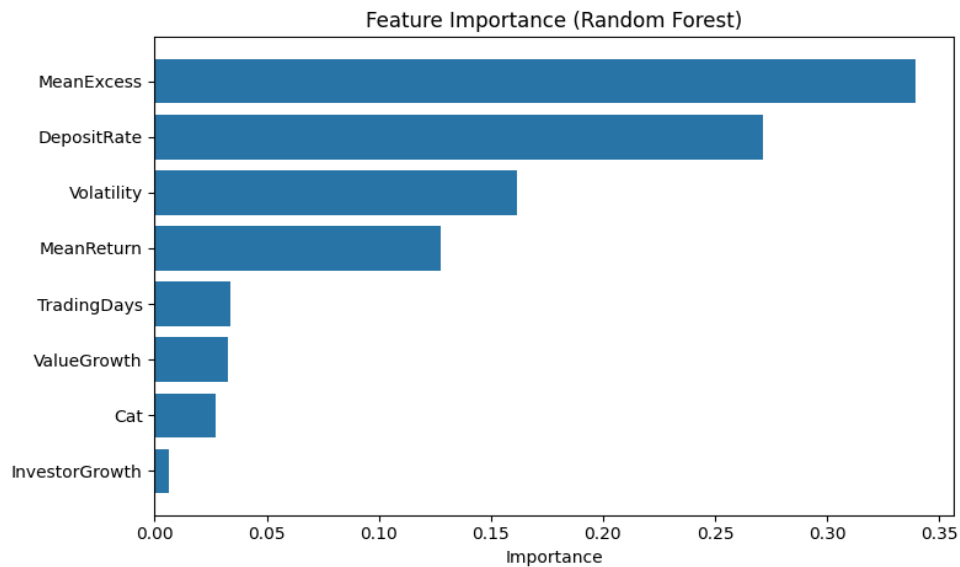


4.3 Feature Importance

The most important features from Random Forest were:

- **MeanExcess (34%)** - if a fund is beating the deposit this month, it tends to keep doing it next month (momentum effect)
- **DepositRate (27%)** - the macro environment matters a lot. When rates are very high, it is harder for funds to beat them
- **Volatility (16%)** - riskier funds are harder to predict
- **MeanReturn (13%)** - similar to excess but without risk-free adjustment

Investor growth and value growth turned out to be almost irrelevant, which surprised me. I thought fund flows would be a good signal, but apparently price-based features dominate.



5. Findings

- **Most fund categories beat the deposit rate** over 2021-2025, but not all. Participation (Islamic) funds failed to beat it.
- **Money Market funds are the risk-adjusted winner** with a Sharpe ratio of 1.87. They barely ever underperform the deposit (119 out of 120 funds beat it).
- **Gold funds had the highest raw excess return** (~27% p.a. over deposit), driven by TL depreciation and the global gold rally. But they are also volatile.
- **Equity funds look good on paper** (51% annual return) but have the highest volatility, so their Sharpe ratio is actually one of the lowest.
- **Category choice matters** - the chi-square test confirms this with overwhelming statistical significance ($p = 10^{-18}$).
- **Next-month performance is partially predictable** - a Decision Tree achieves 73% accuracy, beating the 57% baseline. The strongest predictor is the current month's excess return (momentum).

6. Limitations and Future Work

Limitations

- My categorisation is based on fund names, which is a rough approach. A few funds might be misclassified.
- I did not account for management fees, taxes, or transaction costs.
- The 2021-2025 period in Türkiye was unusual (very high inflation, dramatic policy changes). Findings might not hold in a more stable environment.
- Survivorship bias: funds shut down before December 2025 are underrepresented.
- The ML model uses a simple random split. A time-based split would be more realistic.

Future Work

- Use proper time-series cross-validation for ML models
- Add more features like inflation, USD/TRY rate, BIST index
- Try more advanced models (XGBoost, neural networks)
- Analyse how the 2023 policy shift specifically affected each category

AI Disclosure: Some code scaffolding and parts of this report were drafted with help from Claude (Anthropic). All analysis decisions, interpretations, and personal observations are my own.